SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Total Marks: 50

Code of Conduct

I G. Amruthal (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Compare Type-1 and Type-2 Hypervisors.

A hypervisor is software that creates and manages virtual machines (VMs).

Feature	Type1- Hypervisor	Type-2 Hypervisor.
Execution	Runs directly on hardware.	Runs on a host operating system.
performance	High performance	Comparatively lower.
Security	Smaller attack surface	Depends on host OS security.
manageability	Suitable for large datacenters	Easy for personal use and testing.
Examples	VMware ESXi, Microsoft HyperV	VMware Workstation Pro, oracle virtualbox.

Analysis: Type-1 hypervisors provide better performance, isolation, and resource management for mission critical applications. Type-2 hypervisors are more suitable for development and testing.

Conclusion: Type-1 hypervisors are the better choice for cloud datacenters because they offer high performance, stronger isolation, and centralized management.

2) Workload Isolation in a cloud DataCenter

Virtualization allows Bank6, healthcare, and student portal workloads to run on the same physical hardware using separate virtual machines.

How Isolation works:

=) Each VM has its own virtual CPU, memory, storage, and operating system.

=) the hypervisor controls resource allocation and access between VMs.

=) Virtual networks and firewalls restrict communication between departments.

Two major attack surfaces:

1) Hypervisor Vulnerabilities: Attackers may exploit the hypervisor to access multiple VMs.

2) Virtual network attacks: Attackers may exploit the by weak network segmentation or insecure VM communication.

Conclusion:

Strong hypervisor security, network segmentation, access control, and regular patching help protect workloads.

VM Host Utilization Analysis

Given Utilization:

Parameter	Value	Analysis
CPU	92%	Very high CPU usage.
memory	88%	High memory pressure.
Storage I/O latency	35 ms	Possible storage bottleneck.
VM boot time	120 s	Slow startup

Likely bottlenecks:

CPU: Too many workloads or CPU-intensive applications are competing for processing time.

memory: Insufficient RAM may cause swapping and reduce performance.

Storage: High I/O latency indicates slow disk operations or storage contention.

VM boot time: CPU, memory, storage, or startup service delays may be affecting boot performance.

Corrective actions:

1) Add CPU capacity or migrate VMs to less-loaded hosts.

2) Increase RAM and reduce unnecessary memory usage.

Conclusion: CPU and memory are heavily utilized while storage latency is a likely bottleneck. Resource optimization and capacity upgrades should improve VM boot time and overall performance.

4. Software-defined Datacenter (SDDC).

An SDDC uses software to control and automate computing, storage, and networking resources instead of relying mainly on manual hardware configuration.

Component	Virtualization	Benefit.
Compute	virtual machines and virtual CPUs.	Quick VM creation and scaling.
Storage	Software-defined storage	Flexible capacity and easier data management
Network	Software-defined networking (SDN)	Fast network configuration and segmentation.

Advantages over a traditional datacenter:
Agility, Automation, Scalability, Cost efficiency and flexibility.

Conclusion:
SDDC improves agility through automation, centralized management, and flexible virtualization of compute, storage, and network resources.

Multi-cloud Federation and Identity mismatches.

multi-cloud federation allows applications and users to access resources across multiple cloud providers. Identity mismatches occur when providers use different user identities, roles, authentication methods, or access policies.

problems:

=) Users may have different identities across providers

=) Roles and permission may not match.

=) Weak identity mapping can allow unauthorized access.

proposed secure design:

1) Use a trusted identity provider (IdP) for centralized authentication.

2) Apply federation standards such as SAML 2.0 or OpenID Connect (OIDC)

3) Use OAuth 2.0 where appropriate for delegated authorization

4) map user identities and roles consistently across providers.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :